

AP Precalculus: 2026–2027

Unit 4: Functions Involving Parameters, Vectors, and Matrices Lesson 4.4: Parametrically Defined Circles and Lines (2 instructional periods, 55 minutes each)

Primary Objective: Students will express circular paths parametrically using transformations of $(x(t), y(t)) = (\cos t, \sin t)$ and model linear motion along a line segment using an initial point and constant rates of change.
(*Big Idea: Functions*)

Priority Ideas & Skills

AP Skills

1.B *Express in equivalent forms* — Rewrite a verbal or geometric description of circular or linear motion as a pair of parametric equations, and identify the radius, center, direction, and speed from the algebraic form.

1.C *Construct new functions via transformations* — Apply horizontal/vertical shifts and scale factors to $(\cos t, \sin t)$ to obtain the parametrization of any circle; modify the argument of t to control direction and speed.

Key Understandings (EKs)

EK 4.4.A.1: A complete counterclockwise revolution around the unit circle that starts and ends at $(1, 0)$, centered at the origin, is modeled by $(x(t), y(t)) = (\cos t, \sin t)$ with domain $0 \leq t \leq 2\pi$.

EK 4.4.A.2: Transformations of $(\cos t, \sin t)$ can model any circular path: scaling r sets the radius, shifting by (h, k) moves the center, and replacing t with $-t$ reverses orientation.

EK 4.4.A.3: A linear path along the segment from (x_1, y_1) to (x_2, y_2) can be parametrized by $x(t) = x_1 + (x_2 - x_1)t$, $y(t) = y_1 + (y_2 - y_1)t$ for $0 \leq t \leq 1$, using an initial position and constant rates of change.

Vocabulary, Concepts, & Theorems

Term

Definition / Note

unit circle parametrization	$(x(t), y(t)) = (\cos t, \sin t)$, $0 \leq t \leq 2\pi$; models one full counterclockwise revolution around the unit circle.
radius / center	In $(h + r \cos t, k + r \sin t)$, r is the radius and (h, k) is the center of the circular path.
orientation	The direction of travel along a curve: <i>counterclockwise</i> when t increases in $(\cos t, \sin t)$; <i>clockwise</i> when t is replaced by $-t$.
linear parametrization	$x(t) = x_1 + (x_2 - x_1)t$, $y(t) = y_1 + (y_2 - y_1)t$, $0 \leq t \leq 1$; traces the segment from (x_1, y_1) to (x_2, y_2) at constant speed. The coefficients $(x_2 - x_1)$ and $(y_2 - y_1)$ are the rates of change of x and y with respect to t .

Activate Prior Knowledge & Spiral Review

Spiral topics activated during Warm-Up (first 8 minutes):

- **Cosine and sine values:** $\cos$ and $\sin$ at $0, \pi/2, \pi, 3\pi/2, 2\pi$ — the key outputs that define the unit circle path.
- **Function transformations:** vertical/horizontal shifts and scale factors applied to sine and cosine — maps directly to the $(h + r \cos t, k + r \sin t)$ template.
- **Counterclockwise motion on the unit circle:** interpreting quarter-turn and half-turn positions geometrically — connects the angle t to a physical location on the circle.

Hook — Entry Event

Scenario: A Ferris wheel has a radius of 30 ft and its center is 35 ft above the ground. A rider boards at the rightmost seat, which starts at ground height + 35 ft, exactly 30 ft to the right of center.

Discussion prompts:

1. “Write parametric equations that give the rider’s height y and horizontal position x at time t as the wheel completes one counterclockwise revolution.”
2. “What if the wheel runs clockwise instead? How do you change the equations?”
3. “What is the rider’s highest point, and what value of t achieves it?”

Key tension: A description like “height varies as a sine wave” captures y alone. The parametric form tracks both coordinates simultaneously and encodes direction — essential for real motion.

Lesson — Instruction Sequence

Step 1 — Unit Circle Parametrization (8 min)

Write $(x(t), y(t)) = (\cos t, \sin t)$, domain $0 \leq t \leq 2\pi$. Evaluate at $t = 0, \pi/2, \pi, 3\pi/2, 2\pi$ to fill a class table. Identify the starting point $(1, 0)$, the quarter-turn point $(0, 1)$, the half-turn point $(-1, 0)$, and the return to $(1, 0)$. Emphasize: *counterclockwise* direction, one full revolution. (EK 4.4.A.1)

Step 3 — Linear Parametrization (10 min)

Segment from (x_1, y_1) to (x_2, y_2) :

$$x(t) = x_1 + (x_2 - x_1)t, \quad y(t) = y_1 + (y_2 - y_1)t, \quad 0 \leq t \leq 1.$$

At $t = 0$: start point. At $t = 1$: end point. At $t = 0.5$: midpoint. The coefficients are rates of change: $\Delta x / \Delta t$ and $\Delta y / \Delta t$. Connect to slope: $\Delta y / \Delta x = (y_2 - y_1) / (x_2 - x_1)$. (EK 4.4.A.3)

Step 2 — Transformations of the Circle (12 min)

General form: $x(t) = h + r \cos t$, $y(t) = k + r \sin t$. Show how each parameter changes the picture:

- r scales the radius.
- (h, k) shifts the center.
- Replacing t with $-t$: $\cos(-t) = \cos t$, $\sin(-t) = -\sin t$ — reverses orientation to *clockwise*.
- Replacing t with ct ($c > 1$) speeds up the revolution; replacing with $t/(2\pi) \cdot T$ scales the period to T .

Apply to Ferris wheel: $h = 0$, $k = 35$, $r = 30$. (EK 4.4.A.2)

Step 4 — Apply Both Contexts (10 min)

Circular: Ferris wheel — state equations, find highest point (y is max when $\sin t = 1$, i.e. $t = \pi/2$; height = 65 ft). For one revolution in 40 seconds, scale: $t_{\text{real}} = 40t/(2\pi)$, so rider is highest at $t_{\text{real}} = 10$ seconds.

Linear: Delivery drone flies from $(2, 3)$ to $(8, 11)$. Equations: $x(t) = 2 + 6t$, $y(t) = 3 + 8t$, $0 \leq t \leq 1$. Find position at $t = 0.5$: $(5, 7)$ — the midpoint.

Pacing note: Steps 1–2 fill Period 1; Steps 3–4 + activity in Period 2.

Active Monitoring

- **Radius vs. center confusion:** Students may swap r and (h, k) in the formula. Prompt: “Which part of $h + r \cos t$ shifts the whole circle, and which part controls how wide it is?”
- **Clockwise sign error:** Students may negate both components. Prompt: “What is $\cos(-t)$? What is $\sin(-t)$? So which one changes sign when we reverse direction?”
- **Linear endpoints:** Students may evaluate $t = 0$ and $t = 1$ and get the wrong direction. Ask: “Where does your curve start? Where does it end? Does that match the problem?”
- **Highest point on a circle:** Students may guess the center height. Prompt: “What value of $\sin t$ is the largest possible? What t gives that value?”

Group Work & Differentiation

Students work in groups of 3–4 on the tiered activity.

- **Tier R (Remediate):** Write the parametrization for a circle centered at the origin with radius 5 starting at $(5, 0)$. Verify that the point at $t = \pi/2$ is $(0, 5)$.
- **Tier A (Approaching Proficiency):** Write the parametrization for a circle centered at $(-2, 4)$ with radius 3 traversed *clockwise*. Also write the parametrization for the segment from $(1, 2)$ to $(7, -4)$.
- **Tier E (Extension):** For the Ferris wheel, determine the parameter value t when the rider is at the highest point. If the wheel completes one revolution in 40 seconds, at what actual time (in seconds) is the rider highest? Generalize: if the period is T seconds, write a formula for t_{real} as a function of T .

Individual Work & Assessment

Exit ticket administered independently (final 5 minutes of Period 2).

1. A circle is centered at $(3, -1)$ with radius 4. Write parametric equations for counterclockwise motion starting at $(7, -1)$.
2. Write a parametrization for the line segment from $(0, 6)$ to $(4, 0)$. At what parameter value t does the segment cross the x -axis?

Data use: Students who cannot write the Ferris-wheel equations without scaffolding need more support with the transformation template; revisit in the opening of Lesson 4.5.

Reinforcement & Extension

Homework (Lesson 4.4): 5–6 problems covering: writing circular parametrizations from a description, evaluating at specific t -values, finding when a circular path reaches a given point, writing and evaluating a linear segment parametrization, and one AP-style multiple-choice item.

Preview of Lesson 4.5 — Implicitly Defined Functions: Ask students: “We wrote $x(t) = h + r \cos t$ and $y(t) = k + r \sin t$. If you eliminate t using the identity $\cos^2 t + \sin^2 t = 1$, what equation in x and y do you get?” Lesson 4.5 explores how to recover the Cartesian equation from a parametric form and what is gained or lost.